Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 1 (Generated practice paper - Set 39)

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

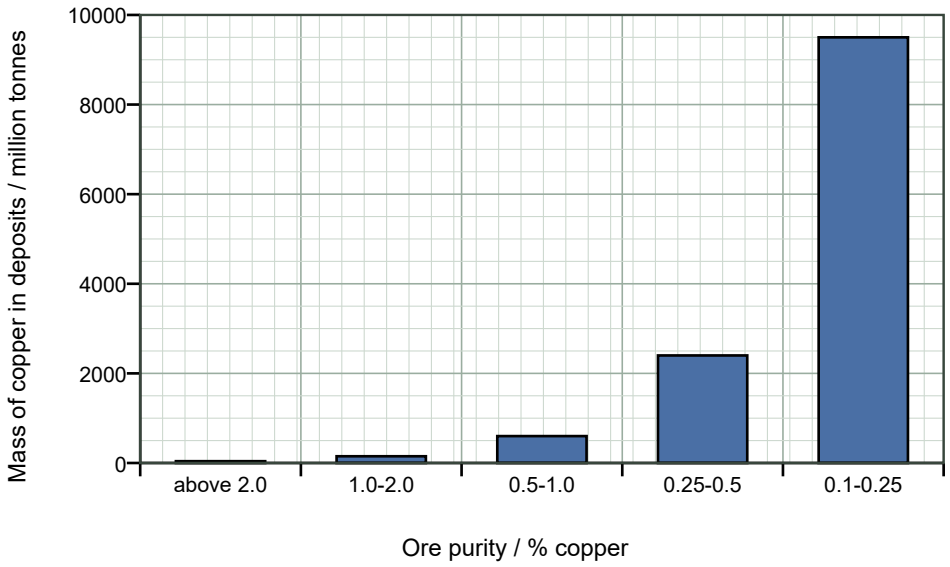
Question	Mark
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11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 Figure 1 shows the mass of copper contained in the Earth's crust at different ore purities, illustrating Lasky's principle.

Figure 1



0 1 . 1 State Lasky's principle.

[1 mark]

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0 1 . 2 Use Figure 1 to calculate how many times more copper is contained in deposits of 0.25-0.5% purity than in deposits above 2.0%.

[1 mark]

Answer ..... times

0 1 . 3 Explain the difference between the **reserves** and the **resource** of a mineral.

[3 marks]

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0 1 . 4 Explain, using Lasky's principle, why the ability to exploit slightly lower grade ores greatly increases reserves.

[2 marks]

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0 1 . 5 Explain why the main problem for future mineral supplies is not the quantity that exists in the crust.

[3 marks]

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0 2 . 1 Which of the following best describes the **precautionary principle**? Tick **one** box.

[1 mark]

A A waste is assumed to cause pollution until research shows it is unlikely to do so ☒

B The polluter must pay for any damage caused ☐

C Emissions should be as low as reasonably achievable ☐

D Only pollutants known to be toxic need to be controlled ☐

0 2 . 2 Explain why the polluter pays principle works best when penalties are high and enforcement is reliable.

[2 marks]

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0 2 . 3 Give **two** choices about emission location that reduce the severity of pollution.

[2 marks]

1 .....

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2 .....

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0 3 . 1 Describe how an **electrostatic precipitator** removes smoke particles from the effluent gases of a coal-fired power station.

[3 marks]

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0 3 . 2 Explain how a **cyclone separator** works.

[2 marks]

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0 3 . 3 Explain why heating coal to remove tar reduces smoke emissions.

[1 mark]

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0 3 . 4 Explain why turbochargers reduce the smoke produced by diesel engines.

[2 marks]

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0 3 . 5 Explain why smaller smoke particles (PM1) are a greater health risk than larger ones.

[2 marks]

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0 4 Table 1 shows some properties of four groups of insecticide.

Table 1

Insecticide group	Toxicity to insects	Toxicity to mammals	Persistence	Liposolubility
Organochlorines (eg DDT)	High	Low	High	High
Organophosphates (eg malathion)	High	High	Low	Low
Pyrethroids	High	Low	Low	Low
Neonicotinoids	High	Low	Moderate	Low (water soluble)

0 4 . 1 Use Table 1 to explain why organochlorine insecticides caused the decline of birds of prey such as the peregrine falcon in the 1950s and 1960s.

[4 marks]

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0 4 . 2 Organophosphates replaced organochlorines for many uses. Use Table 1 to give **one** advantage and **one** disadvantage of this change.

[2 marks]

Advantage .....

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Disadvantage .....

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0 4 . 3 Explain why pyrethroids should not be used near rivers or fish farms.

[1 mark]

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0 4 . 4 Describe how the properties of neonicotinoids allow them to be used as a seed coating, and explain why this method of application reduces harm to non-target species.

[3 marks]

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0 5 A 1960s office block is being refurbished. Table 1 shows the measured heat loss through its fabric before the work.

Table 1

Part of the building	Heat loss / kW
Walls (uninsulated)	42
Roof	26
Single-glazed windows	31
Floor	9
Draughts and ventilation	22

0 5 . 1 Calculate the percentage of the total heat loss that passes through the windows.

[2 marks]

Answer ..... %

0 5 . 2 Explain how each of the following reduces heat loss.

[3 marks]

Cavity wall insulation .....

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Low-emissivity double glazing .....

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A heat exchanger on the ventilation system .....

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0 5 . 3 Explain why there is an optimum thickness of loft insulation rather than simply using as much as possible.

[2 marks]

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0 5 . 4 The refurbishment also includes a combined heat and power (CHP) unit. Explain why a CHP unit has a lower **electrical** efficiency than a large power station but a much higher **overall** efficiency.

[3 marks]

0 5 . 5 Explain **five** other ways the refurbished building could be designed or managed to use less energy.

[5 marks]

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0 6 . 1 Which gas makes up about 21% of the dry atmosphere by volume? Tick **one** box.

[1 mark]

A Nitrogen ☐

B Oxygen ☐

C Argon ☐

D Carbon dioxide ☐

0 6 . 2 Explain why it is difficult to predict the effect of a human activity on atmospheric processes.

[2 marks]

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0 6 . 3 Explain how winds transport both energy and water to areas such as the UK.

[2 marks]

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0 7 A group of students wants to investigate the effect of vegetation cover on rain-splash soil erosion.

0 7 . 1 Describe how they could carry out the investigation in the laboratory.

[5 marks]

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0 7 . 2 Give **three** variables, other than those you have standardised, that affect rain-splash erosion in the field.

[3 marks]

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0 7 . 3 Explain why it is easier to obtain valid results in the laboratory than in the field, and give **one** disadvantage of laboratory studies.

[2 marks]

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0 8 . 1 Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

A High sand content ☐

B High clay content ☐

C Crumb peds ☐

D High pH ☐

0 8 . 2 Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

0 8 . 3 Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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0 9 Table 1 shows the approximate energy density of some energy resources.

Table 1

Energy resource	Energy density / MJ kg^{-1}
Uranium (natural, in fission reactor)	500 000
Natural gas	55
Petrol	46
Coal	28
Dry wood	16
Lithium-ion battery (stored electricity)	0.7

0 9 . 1 Explain what is meant by **energy density** and why it is an important feature of an energy resource.

[3 marks]

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0 9 . 2 Use Table 1 to calculate how many tonnes of coal contain the same energy as 1 kg of uranium.

[2 marks]

Answer ☐ tonnes

0 9 . 3 Explain why low energy density does not prevent solar and wind power from being useful.

[2 marks]

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0 9 . 4 Explain the difference between an **intermittent** resource and an **unpredictable** resource, using named examples.

[2 marks]

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0 9 . 5 Explain why natural gas must be liquefied for transport by ship.

[1 mark]

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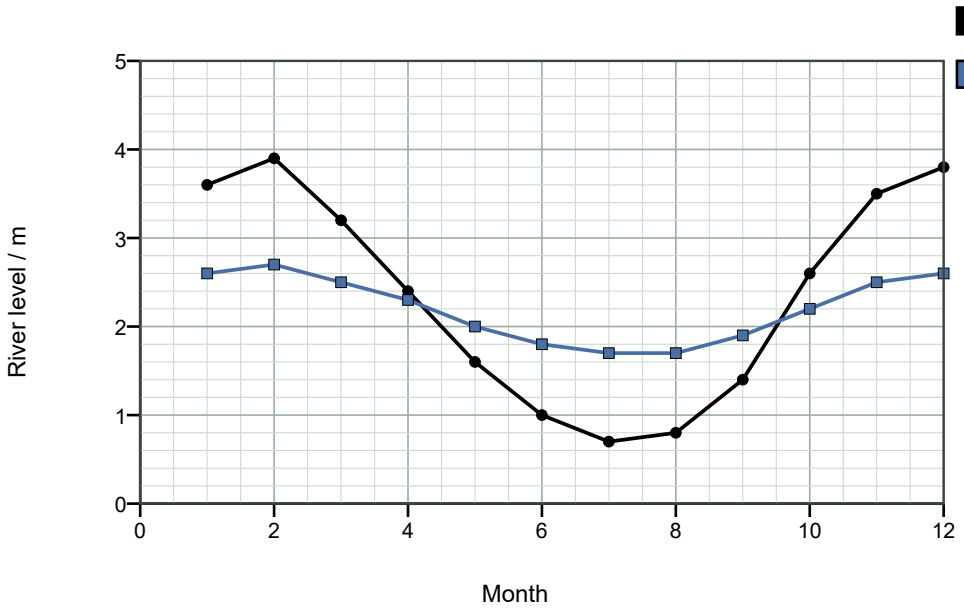
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1 0 Figure 1 shows the river level below a reservoir site during one year, before and after a river-regulation reservoir was built.

Figure 1



1 0 . 1 Use Figure 1 to calculate the reduction in the annual range of river level produced by the reservoir.

[2 marks]

Answer ..... m

1 0 . 2 Explain how the reservoir produces the changes shown in Figure 1.

[2 marks]

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1 0 . 3 Explain why human activities in the catchment may have increased the extremes in river flow before the reservoir was built.

[2 marks]

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1 0 . 4 Discuss the factors that should be considered when selecting a site for a new reservoir, and evaluate the alternative strategies for increasing the sustainability of water supplies.

[9 marks]

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Extra space

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Write an essay on one of the following topics.

1 1 . 1 Discuss the strategies that can be used to provide sustainable supplies of water for a growing population.

[25 marks]

OR

1 1 . 2 Evaluate the contribution of new technologies to the monitoring and understanding of environmental change in the physical environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS